NINTH ANNUAL ECC
UNDERGRADUATE
MATHEMATICS COMPETITION

April 1, 2006

Time control: Three hours

NO BOOKS, NOTES, CALCULATORS, COMPUTERS OR NON-TEAM-MEMBERS may be consulted.

PLEASE BEGIN EACH PROBLEM ON A NEW SHEET OF PAPER. Team identification and problem number should be clearly given at the top of each sheet of paper submitted.

Each problem counts 10 points. Partial credit for significant but incomplete work. For full credit, answers must be fully justified. But this may in some cases simply mean showing all work and reasoning. Have fun!

1. Roots of a quadratic.

Given that $3+2i$ is one of the roots of the quadratic equation $2x^2+rx+s=0$, where r and s are real numbers, find the value of s.

2. Functional equation.

Suppose that f, g, h and k are functions and a and b are real numbers satisfying

$$f(x) = (x-2)g(x)+1 = (x+2)h(x)+5 = (x^2-4)k(x)+ax+b$$

for all real x. Determine a and b.

3. Square root of an integer.

Prove that $\sqrt{5}+\sqrt{11-2\sqrt{30}}$ is the square root of an integer.

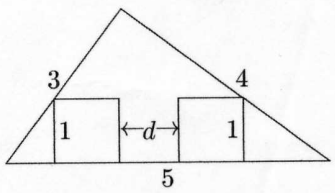
4. Limit of quotient.

Evaluate

$$\lim_{h\to 0} \frac{\int_2^{2+h} e^{(x^4-1)}dx}{h}.$$

Justify your answer.

5. Distance between squares.



Two unit squares stand on the hypotenuse of a $(3,4,5)$ triangle in such a way that they lie inside the triangle, and a corner of one touches the side of length 3 and a corner of the other touches the side of length 4, as shown in the figure at the left. What is the distance d between the squares?

6. Which integral is larger?

Determine which of the following integrals has the larger value:

$$I_1 = \int_0^1 \cos x^2 dx, \qquad \text{or} \quad I_2 = \int_0^1 \cos \sqrt{x} dx$$

Justify your answer.

7. Divisible by 33.

Prove that

$$5^{2n+1} + 11^{2n+1} + 17^{2n+1}$$

is divisible by 33 for every integer $n \geq 0$.

8. Urns and balls.

An urn contains twenty-one balls, each bearing one of the numbers $\{1, 2, 3, 4, 5, 6\}$. One ball is numbered 1, two balls bear the number 2, three balls bear number 3, four bear number 4, five bear number 5, and six bear number 6. If two balls are drawn at random (without replacement), what is the probability that the sum of the two numbers on them is more than 6?

9. A sum less than 1.

Show that

$$\frac{1}{2\sqrt{1}+1\sqrt{2}} + \frac{1}{3\sqrt{2}+2\sqrt{3}} + \frac{1}{4\sqrt{3}+3\sqrt{4}} + \cdots + \frac{1}{2006\sqrt{2005}+2005\sqrt{2006}} < 1.$$

10. Differential inequalities.

Prove that there is no polynomial function $f(x)$ which satisfies both

$$f(x) > f''(x) \quad \text{for all } x$$

and

$$f'(x) > f''(x) \quad \text{for all } x.$$